

GROUP TRAVEL EXPENSE SIMPLIFICATION ENGINE

Lab 1: Requirements Engineering & UML Use-Case Modelling

PES University – Department of CSE

Problem Statement #37 | Retail, E-Commerce & Finance

1. Requirements Specification

1.1 Problem Context & Overview

A multi-currency expense sharing utility that records uneven group expenses, converts foreign currencies via daily exchange rates, and computes minimum-transaction debt settlement graphs.

Target Stakeholders / Actors: Group Traveler, Trip Admin

1.2 Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record a group expense with payer, amount, currency, participants, and each participant's share.	High	Pass: A saved expense contains the payer, positive amount, currency, participants, and valid shares whose total equals the expense amount. Fail: Any required field is missing or the shares do not reconcile to the expense amount.	Accurate expense capture is necessary to calculate each traveler's financial responsibility.
FR-002	Functional	The system shall convert recorded expenses between supported currencies using the applicable daily exchange rate.	High	Pass: A foreign-currency expense is converted using the exchange rate for the selected transaction date and the converted value is displayed. Fail: The required rate is unavailable or the conversion is inconsistent with the stored rate.	Multi-currency trips require a common valuation basis for reliable balances.
FR-003	Functional	The system shall calculate and display each traveler's net balance for a trip from all recorded expenses.	High	Pass: For every traveler, total paid minus total owed is calculated correctly and balances reconcile to zero across the group. Fail: Any traveler's balance is incorrect or the group balances do not net to zero.	Net balances provide a clear view of who should receive or pay money.
FR-004	Functional	The system shall compute a minimum-transaction debt settlement graph that minimizes the total number of peer-to-peer payments required to settle the group.	High	Pass: The generated settlement plan preserves every traveler's net balance and uses a minimum or algorithmically minimized number of payment edges. Fail: The plan does not reconcile balances or contains unnecessary payment edges relative to the selected algorithm.	Minimizing transactions makes final settlement faster and easier for travelers.
FR-005	Functional	The system shall allow authorized users to add, edit, and remove trip expenses and participants while preserving an auditable record of changes.	Medium	Pass: An authorized user can perform the permitted operation and the system records who changed what and when. Fail: An unauthorized user changes protected trip data or an audit entry is missing.	Travel plans change frequently, so corrections must be possible without losing accountability.

1.3 Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	The core balance and settlement calculation engine shall execute in under 100 ms for groups of up to 50 travelers and 500 recorded expenses under normal operating conditions.	High	Pass: Benchmark tests meet the 100 ms target for the stated workload under normal operating conditions. Fail: The measured execution time exceeds 100 ms for the defined workload.	Fast calculations keep the expense-sharing experience responsive as trips grow.
NFR-002	Security	The system shall protect trip and financial data through authenticated access, role-based authorization, encrypted communication, and secure storage of sensitive information.	High	Pass: Protected operations require authentication, role permissions are enforced, network communication uses HTTPS/TLS, and sensitive data is stored using approved security controls. Fail: Unauthenticated or unauthorized users can access or modify protected financial or trip data.	Group expense data contains financial information and should only be accessible to permitted users.

2. UML Use-Case Diagram

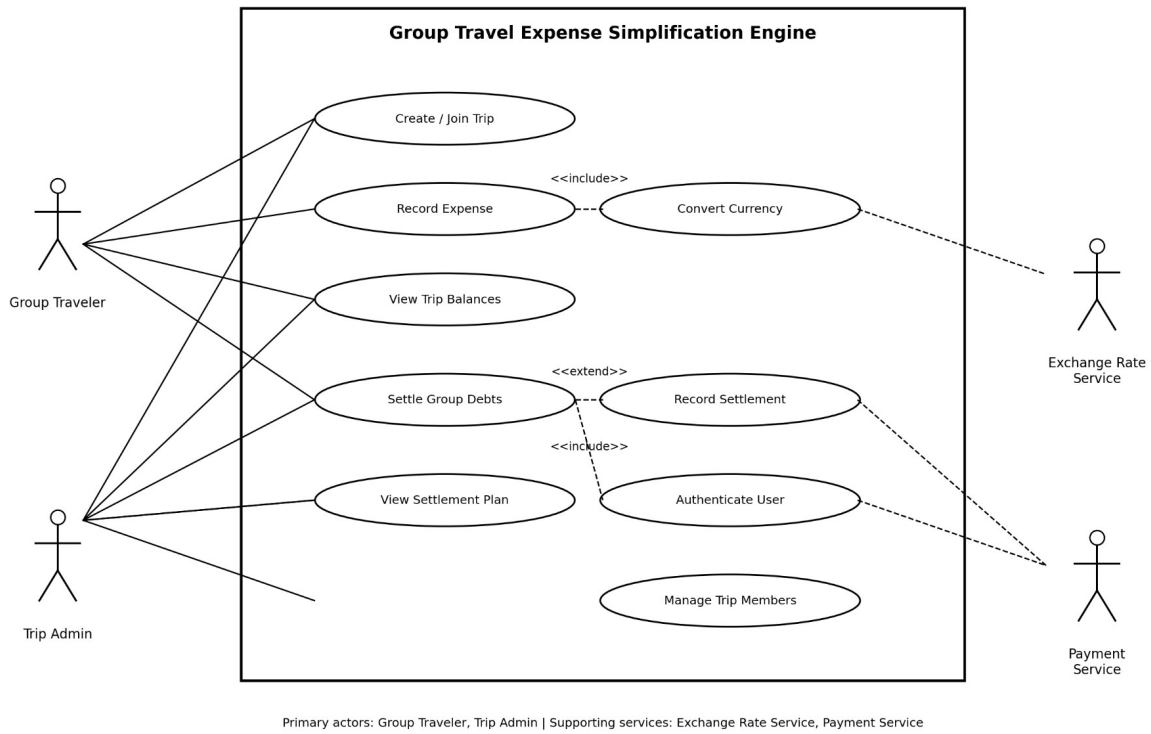


Figure 1. Group Travel Expense Simplification Engine – UML Use-Case Diagram

3. Use-Case Flow Specification

Use Case ID	UC-001
Use Case Name	Settle Group Debts
Primary Actor	Group Traveler
Supporting Actor	Trip Admin
Goal	Generate and optionally confirm a minimum-transaction payment plan that settles all trip balances.

3.1 Preconditions

1. The traveler is authenticated and has access to the trip.
2. The trip contains at least two travelers.
3. Recorded expenses are valid and the latest balances have been calculated.
4. All required exchange-rate conversions for the trip's recorded expenses are available.

3.2 Main Success Scenario

5. Group Traveler selects Settle Group Debts for the trip.
6. The system retrieves the latest recorded expenses and current net balances for all travelers.
7. The system verifies that the group balances reconcile to zero.
8. The system separates travelers with positive balances from travelers with negative balances.
9. The system computes a debt settlement graph that minimizes the number of peer-to-peer payment transactions.
10. The system displays each proposed payment with payer, recipient, and amount.
11. The Group Traveler reviews the proposed settlement plan.
12. The Group Traveler confirms the plan.
13. The system records the confirmed settlement instructions and marks the relevant balances as pending settlement.
14. The use case ends successfully.

3.3 Alternate Flow – Balance or Exchange-Rate Problem

15. The system detects that the group balances do not reconcile to zero, or a required exchange rate is unavailable.
16. The system does not generate a settlement plan.
17. The system identifies the affected expense or conversion data and displays an error message.
18. The user corrects the expense or waits for the required exchange-rate data to become available.
19. The user retries Settle Group Debts after the data is valid.

3.4 Postconditions

20. A valid settlement graph is displayed and, if confirmed, stored for the trip.
21. The settlement instructions preserve every traveler's calculated net balance.
22. The system records the confirmation and settlement status for audit purposes.

3.5 Success Guarantee

The system provides a valid, transaction-efficient payment plan that settles the group's current net balances without changing the underlying expense records.